Alibaba Cloud

Customer portal 21

Architecture proposal

Prepared for Acme Pay Pte Ltd

Date	Revisions	Products	Catalog checked
2 November 2026	4	21	2026-10-31

Prepared with Architecture Builder

AI-generated. Review with a solutions architect before you build.

Executive summary

Acme Pay Pte Ltd is building a customer portal that must satisfy two non-negotiable demands: strict Singapore data residency and a fast, reliable experience for users worldwide. This proposal translates those requirements into a production-ready architecture on Alibaba Cloud, designed specifically for a regulated fintech workload.

The design spans high-availability compute, encrypted storage, automated disaster recovery, and an AI-powered support chatbot—all pinned to the Singapore region. For your European customers who reported slow load times, we introduce private backbone routing that accelerates their experience without moving a single byte of data outside Singapore.

For your sales and leadership teams, the business value is straightforward: lower operational risk, faster time-to-market for new features like AI support, and a consistent customer experience that protects your brand reputation across geographies. Every component below exists to remove a specific risk or unlock measurable value for your platform.

What this design delivers

Regulatory confidence and risk reduction	By pinning all compute, storage, encryption, and AI services to Singapore, the architecture provides a verifiable boundary for auditors. This eliminates the legal and financial risks of accidental data export, allowing your compliance team to approve the platform faster.
Resilient revenue operations	The multi-zone deployment combined with automated backups means the portal can survive hardware failures or localized outages without manual intervention. For a payment platform, this uptime directly protects transaction volume and prevents costly SLA breaches.
Global reach without compromise	European customers will experience significantly faster load times through private backbone acceleration. This improves conversion rates and user satisfaction in a key market, proving that strict data residency does not have to mean a degraded international user experience.

What we propose next

1. Schedule a technical deep-dive with your engineering leads to review the network topology and failover mechanics.

2. Conduct a data residency mapping exercise with your compliance officer to formally sign off on the Singapore-only deployment footprint.

What you told us, and how the design answers it

Each priority from our conversation, where it came from, and the part of the design that answers it.

All customer data, including documents, database records, and AI processing, must remain physically within Singapore to meet regulatory obligations.

"customer data must stay in Singapore"

How the design answers it

Every service in this architecture—compute, databases, object storage, vector search, and AI inference—is deployed exclusively in the Singapore region. Encryption keys are managed locally. This ensures you can demonstrate full compliance to regulators without architectural compromises, avoiding the severe fines and reputational damage associated with cross-border data leaks.

Products involved

Elastic Compute Service (ECS), ApsaraDB RDS, Object Storage Service (OSS), Key Management Service (KMS), Vector Retrieval Service for Milvus, Alibaba Cloud Model Studio, Cloud Backup

The portal requires a robust disaster recovery strategy to ensure business continuity during infrastructure failures.

"Add disaster recovery."

How the design answers it

We deploy your primary database and application servers across two distinct availability zones in Singapore, supplemented by a cross-zone read replica and automated backups. If a zone fails, traffic shifts seamlessly. Without this, a single facility outage could take your payment portal offline, directly impacting revenue and customer trust.

Products involved

ApsaraDB RDS, Cloud Backup, Elastic Compute Service (ECS), Application Load Balancer (ALB)

An intelligent self-service channel is needed to answer customer questions instantly, reducing support costs.

"Add an AI chatbot that answers customer questions."

How the design answers it

By combining a managed vector store with large language model inference, the portal can retrieve answers directly from your own documentation. This deflects routine queries from human agents, lowering operational costs while providing 24/7 support. All retrieval and generation happens in Singapore, keeping proprietary knowledge secure.

Products involved

Vector Retrieval Service for Milvus, Alibaba Cloud Model Studio, Elastic Compute Service (ECS)

European users require acceptable page load speeds to prevent churn and maintain a competitive edge.

“Our users in Europe say the portal is slow. What can we do?”

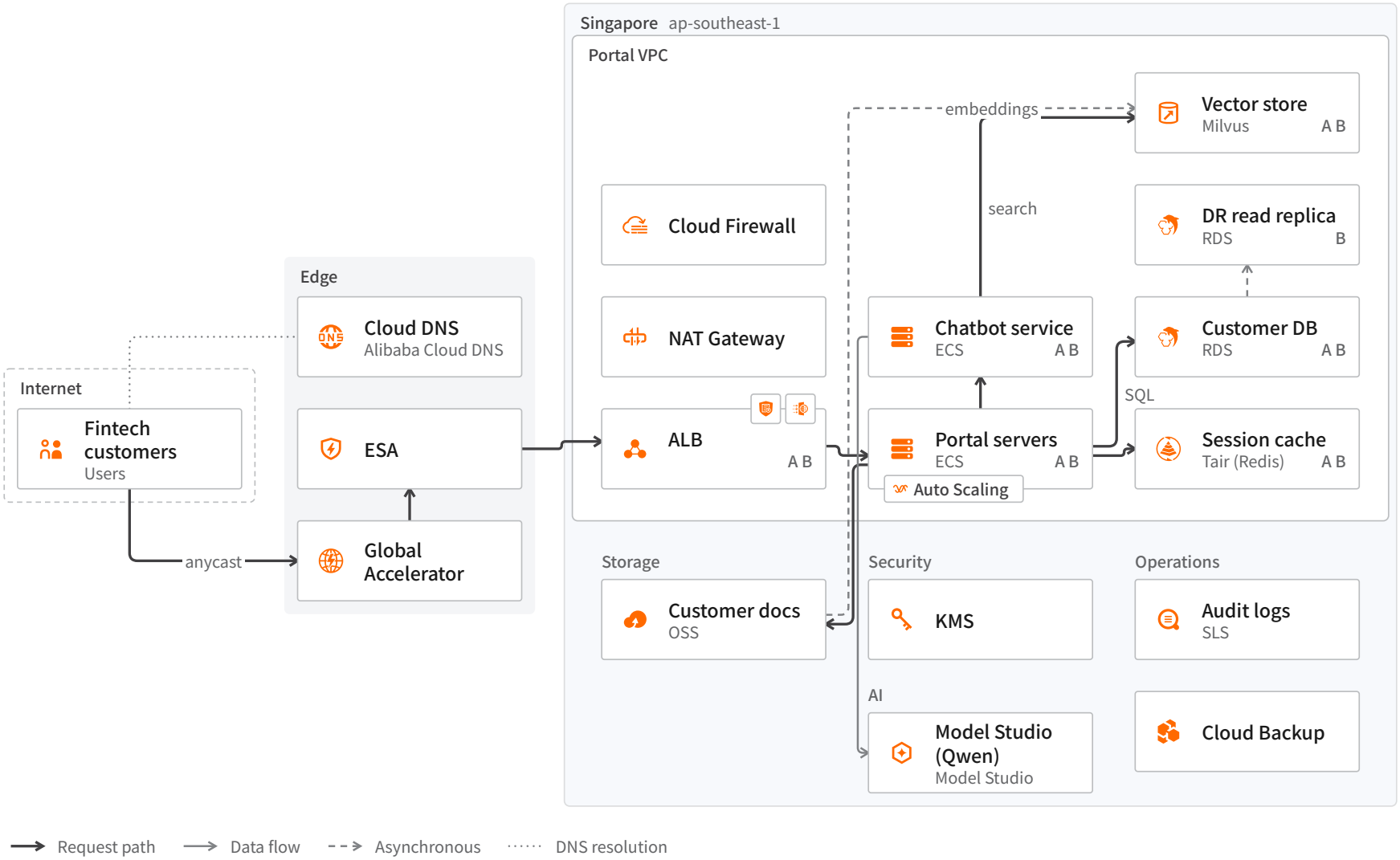
How the design answers it

Rather than replicating data in Europe—which would violate your residency rules—we route European traffic over Alibaba Cloud's private global backbone directly to Singapore. This dramatically reduces latency and packet loss compared to the public internet, giving European users a local-like experience while your compliance posture remains untouched.

Products involved

Global Accelerator (GA), Edge Security Acceleration (ESA), Alibaba Cloud DNS

Architecture diagram



Components

Every product in the diagram, why it is in this design, and where to read its official documentation.

Users

Fintech customers
Internet

Why it's here
The entry point for your customers; the entire architecture is optimized around their need for a secure, fast login and transaction experience.

What it does
People reaching the system over the internet from browsers or apps.

Alibaba Cloud DNS

Edge · Domain Names and Websites

Why it's here
Translates your portal's domain name reliably. Without it, users and the acceleration layer cannot find your application, causing immediate downtime.

What it does
Authoritative public DNS for your domains; a resolve hop, not on the data path.

Official documentation
www.alibabacloud.com/product/dns

Edge Security Acceleration (ESA)

Edge · Networking & CDN

Why it's here
Absorbs distributed denial-of-service attacks and caches static content at the edge. Leaving this out exposes a financial application to extortion-level outages and slower page loads.

What it does
Global edge: caching, dynamic acceleration, WAF, DDoS defense, bot management and edge functions.

Official documentation
www.alibabacloud.com/product/esa

Global Accelerator (GA)

Edge · Networking & CDN

Why it's here
Routes European users over a private network path instead of the congested public internet. This is the specific solution to your reported latency issues, retaining customers who might otherwise abandon a slow portal.

What it does
Anycast entry points carrying traffic over the Alibaba backbone to regional endpoints.

Official documentation
www.alibabacloud.com/product/ga

Cloud Firewall

Singapore (ap-southeast-1) · Portal VPC · Security

Why it's here

Inspects all traffic entering your private network. It acts as the central checkpoint to enforce security policies, preventing unauthorized access to your financial infrastructure.

What it does

Internet, VPC and NAT border firewalls with intrusion prevention and access control.

Official documentation

www.alibabacloud.com/product/cfw

NAT Gateway

Singapore (ap-southeast-1) · Portal VPC · Networking & CDN

Why it's here

Allows your internal servers to download security patches without exposing them directly to the public internet, closing a major attack vector for financial systems.

What it does

Gives private VPC resources outbound internet (SNAT) and mapped inbound ports (DNAT).

Official documentation

www.alibabacloud.com/product/nat

Application Load Balancer (ALB)

Singapore (ap-southeast-1) · Portal VPC · Zones A, B

Why it's here

Distributes incoming user sessions evenly across your servers in two zones. Without it, a failure in one server takes down all connected users rather than just rerouting them.

What it does

Layer-7 HTTP/HTTPS/gRPC load balancer with content routing; also serves as ACK Ingress.

Official documentation

www.alibabacloud.com/product/server-load-balancer

Web Application Firewall (WAF)

WAF 3.0

Singapore (ap-southeast-1) · Portal VPC · On ALB

Why it's here

Blocks common web exploits targeting financial portals. A single successful injection attack could expose customer records, making this essential for fintech risk management.

What it does

WAF 3.0: blocks OWASP attacks, CC floods and bots via cloud-native or CNAME access.

Official documentation

www.alibabacloud.com/product/web-application-firewall

Certificate Management Service

TLS certificate
Singapore (ap-southeast-1) · Portal VPC · On ALB

Why it's here
Manages the encryption certificates for HTTPS. Expired or misconfigured certificates cause browser warnings that immediately destroy customer trust in a payment platform.

What it does
Buy, issue and deploy SSL/TLS certificates, including a private CA.

Official documentation
www.alibabacloud.com/product/certificates

Elastic Compute Service (ECS)

Computing

Why it's here
Provides the core computing power for both the web portal and the chatbot backend. It is sized to run across two zones so no single hardware failure stops your business.

What it does
Virtual machines (x86, Arm, GPU) with cloud disks and security groups.

Official documentation
www.alibabacloud.com/product/ecs

Auto Scaling (ESS)

Singapore (ap-southeast-1) · Portal VPC · On Portal servers

Why it's here
Automatically adds servers during peak login windows and removes them when quiet. This prevents slow-downs during high traffic while ensuring you do not pay for idle capacity overnight.

What it does
Scales ECS or ECI fleets by rules, schedules or metrics and registers them to load balancers.

Official documentation
www.alibabacloud.com/product/auto-scaling

Vector Retrieval Service for Milvus

Vector store
Singapore (ap-southeast-1) · Portal VPC · Zones A, B

Why it's here
Stores the mathematical representations of your support documents so the chatbot can find relevant answers instantly. Without it, the AI has no factual basis to draw from, leading to inaccurate responses.

What it does
Managed vector database compatible with open-source Milvus, for RAG and multimodal search.

Official documentation
www.alibabacloud.com/product/milvus

ApsaraDB RDS
Database

Why it's here
The secure vault for all structured customer and transaction data. The cross-zone replica ensures that if the primary database is corrupted or unreachable, your business continues operating.

What it does
Managed MySQL, PostgreSQL, SQL Server and MariaDB with HA, read replicas and backups.

Official documentation
www.alibabacloud.com/product/apsaradb-rds

Tair (Redis® OSS-Compatible)
Session cache
Singapore (ap-southeast-1) · Portal VPC · Zones A, B

Why it's here
Holds user sessions and security tokens in memory. Without it, every page click would require a database lookup, making the portal unusably slow and increasing database costs.

What it does
Managed Redis-compatible in-memory database and cache, plus Tair enterprise features.

Official documentation
www.alibabacloud.com/product/tair

Object Storage Service (OSS)
Customer docs
Singapore (ap-southeast-1) · regional services · Storage

Why it's here
Provides highly durable, encrypted storage for KYC documents and identity files. Storing these on standard servers risks data loss and fails compliance checks for document retention.

What it does
Durable object storage for files, media, backups and data lakes; S3-compatible.

Official documentation
www.alibabacloud.com/product/object-storage-service

Key Management Service (KMS)
Singapore (ap-southeast-1) · regional services · Security

Why it's here
Controls the encryption keys for your databases and document storage. In a regulated fintech, storing sensitive data without centralized key management is a direct compliance violation.

What it does
Managed encryption keys with rotation and BYOK, plus Secrets Manager.

Official documentation
www.alibabacloud.com/product/kms

Alibaba Cloud Model Studio

Model Studio (Qwen)
Singapore (ap-southeast-1) · regional services · AI & Machine Learning

Why it's here
Powers the natural language understanding for the chatbot. It turns retrieved facts into conversational answers, reducing the burden on your human support staff while keeping processing inside Singapore.

What it does
Managed model APIs for Qwen (such as qwen3.8-max) and third-party models; OpenAI-compatible.

Price
Pay per token; qwen3.8-max $2 per 1M input and $6 per 1M output tokens (Singapore); 1M free tokens per model for 90 days, as of 2026-10-31

Official documentation
www.alibabacloud.com/product/modelstudio

Simple Log Service (SLS)

Audit logs
Singapore (ap-southeast-1) · regional services · Storage

Why it's here
Collects and stores audit logs from every component. Regulators require proof of who accessed what data and when; without centralized logging, passing an audit becomes nearly impossible.

What it does
Collects, searches, analyzes with SQL and alerts on logs, metrics and traces.

Official documentation
www.alibabacloud.com/product/log-service

Cloud Backup

Singapore (ap-southeast-1) · regional services · Storage

Why it's here
Creates scheduled, recoverable copies of your servers, databases, and documents. If ransomware or human error corrupts your live systems, this is your ultimate insurance policy to restore operations.

What it does
Backup and disaster recovery for ECS, databases, NAS, OSS and VMware.

Official documentation
www.alibabacloud.com/product/hybrid-backup-recovery

Considerations and next steps

Trade-offs and how they are handled

Maintaining high availability across two zones with dedicated disaster recovery components (read replicas, backup vaults) increases baseline infrastructure costs compared to a single-server setup.

How it is handled

Auto Scaling (ESS) dynamically adjusts compute resources based on actual demand, ensuring you only pay for peak capacity when necessary, offsetting the fixed costs of resilience.

Routing European traffic through Global Accelerator introduces a per-gigabyte data transfer cost that is higher than standard public internet routing.

How it is handled

The cost is justified by the reduction in customer churn due to slow performance. Edge Security Acceleration (ESA) also caches static assets closer to Europe, reducing the volume of data that must traverse the expensive backbone.

Running an AI chatbot with continuous vector search and LLM inference adds ongoing compute and API consumption costs to the operational budget.

How it is handled

The chatbot answers routine questions that would otherwise reach your support team. Tracking the support hours it saves against Model Studio and Milvus usage shows whether it pays for itself, and usage can be capped while that is measured.

Next steps

1. Schedule a technical deep-dive with your engineering leads to review the network topology and failover mechanics.
2. Conduct a data residency mapping exercise with your compliance officer to formally sign off on the Singapore-only deployment footprint.
3. Execute a proof-of-concept test routing sample European traffic through Global Accelerator to measure real-world latency improvements.
4. Finalize the commercial proposal, including reserved instance pricing for predictable workloads and pay-as-you-go rates for burstable AI usage.

Design record

What the design rests on: the requirements pinned in the conversation, constraints, assumptions and open questions, with the architect's first notes.

Architect's first notes

I'll design a two-zone fintech customer portal in Singapore with ESA, ALB, WAF, ECS, Tair and RDS.

21 products, 22 connections, 4 revisions

Pinned requirements

Customer data must stay in Singapore	Held Pinned in revision 1
Customer data stays in Singapore	Held Pinned in revision 2

Constraints

Data and workloads stay in Singapore (ap-southeast-1)

Assumptions

No assumptions were recorded.

Connections

How the products in the diagram talk to each other.

From	To	Carries	Type
Fintech customers	Cloud DNS		DNS resolution
ESA	ALB	origin	Data
ALB	Portal servers	HTTP	Data
Portal servers	Session cache	Redis	Data
Portal servers	Customer DB	SQL	Data
Customer DB	KMS		Control
Portal servers	NAT Gateway		Control
ALB	Cloud Firewall		Control
Portal servers	Audit logs		Control
Customer DB	Audit logs		Control
Portal servers	Customer docs	objects	Data
Customer DB	DR read replica	replication	Asynchronous
Portal servers	Cloud Backup		Control
Customer DB	Cloud Backup		Control
Customer docs	Cloud Backup		Control
Portal servers	Chatbot service	chat API	Data
Chatbot service	Vector store	search	Data
Chatbot service	Model Studio (Qwen)	LLM API	Data
Customer docs	Vector store	embeddings	Asynchronous
Chatbot service	Cloud Backup		Control
Fintech customers	Global Accelerator	anycast	Data
Global Accelerator	ESA	backbone	Data

Revision history

Every ask and what it changed in the design.

Revision 1

Drawn in 23.0 s

Ask

We're a Singapore fintech and customer data must stay in Singapore. Design a web app with a database for our customer portal.

Changes

+ Users · Fintech customers · in Internet
+ Cloud DNS
+ ESA · in Edge
+ ALB · zones A, B
+ WAF · WAF 3.0 · on ALB
+ ECS · Portal servers · zones A, B
+ Auto Scaling · on ECS
+ Tair (Redis) · Session cache · zones A, B
+ RDS · Customer DB · zones A, B
+ OSS · Customer docs · in Singapore (ap-southeast-1)
+ SLS · Audit logs · in Singapore (ap-southeast-1)
+ KMS · in Singapore (ap-southeast-1)
+ Cloud Firewall · in Portal VPC
+ NAT Gateway · in Portal VPC
+ Certificates · TLS certificate · on ALB
+ 12 connections

Revision 2

Drawn in 9.6 s

Ask

Add disaster recovery.

Changes

+ Cloud Backup · in Singapore (ap-southeast-1)
+ RDS · DR read replica · zones B
+ 4 connections

Revision 3

Drawn in 7.3 s

Ask

Add an AI chatbot that answers customer questions.

Changes

+ ECS · Chatbot service · zones A, B
+ 5 connections
+ Milvus · Vector store · zones A, B
+ Model Studio · Model Studio (Qwen) · in Singapore (ap-southeast-1)

Revision 4

Drawn in 4.5 s

Ask

Our users in Europe say the portal is slow. What can we do?

Changes

+ Global Accelerator
+ 2 connections · Fintech customers to Global Accelerator to ESA
− 1 connection removed · Fintech customers to ESA